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Supplemental information

Superior polymeric gas separation membrane designed by explainable graph machine learning

Jiaxin Xu, Agboola Suleiman, Gang Liu, Michael Perez, Renzheng Zhang, Meng Jiang, Ruilan Guo, and Tengfei Luo

Supplemental Note(s)

Section 1. Machine learning models and performance evaluation

Several popular baseline models in molecular property prediction, both GNN-based and traditional non-GNN-based, are implemented in comparison with GREA. The graph convolutional network (GCN) is a variant of convolutional neural networks which can operate directly on graphs and encode node features efficiently. The graph isomorphism network (GIN) generalizes the Weisfeiler-Lehman isomorphism test and has high discriminative and representational power among GNNs for graph-level tasks. Besides GNN methods, several traditional ML algorithms are also implemented. Random forest (RF) is an ensemble of decision trees usually trained with the "bagging" method and is one of the most used ML algorithms because of its robustness and simplicity[S1]. Gaussian process regression (GPR) is a probabilistic supervised learning model that can incorporate prior knowledge into its kernel functions and provide predictions along with uncertainty estimations[S2]. Multilayer perceptron (MLP) is a feed-forward neural network model composed of multiple layers of interconnected neurons.

Section 2. Dataset and data cleaning

The labeled dataset in this paper is built based on the public Polymer Gas Separation Membrane Database downloaded from the Membrane Society of Australasia (MSA) portal[S3], which hosts a variety of experimental gas permeability data for around 1500 polymers published from 1950 to 2018. Five major gases in the gas separation industry – nitrogen (N_2), oxygen (O_2), hydrogen (H_2), methane (CH_4), and carbon dioxide (CO_2) - are mainly included in the measurements. A wide range of membrane materials is covered including rubber and glassy polymers, carbon molecular sieves, and zeolites. To include more up-to-date high-performance gas separation polymers, we manually collected around 90 experimental gas permeability data from 19 studies in the literature published between 2018 and 2022, with a special focus on polyimides. SMILES (Simplified Molecular Input Line Entry System) is a widely used string line notation in the chemistry community for uniquely describing the fundamental chemical components and structures of different molecules[S4]. Polymer-SMILES (p-SMILES) is a specific class of SMILES notation for polymers with "*" denoting the polymerization points of monomers[S5]. For polymer structure representation, we manually added the corresponding p-SMILES strings to every entry of the dataset and removed those entries that (a) cannot be represented as a plain p-SMILES form (e.g., composite materials, doped and modified materials, or materials with no specific repeating units and structures stated in the literature) and (b) are copolymers (herein only homopolymers are considered for the simplicity of polymer representation). Regarding a single polymer structure with multiple experimental values recorded, produced by different synthesizing and testing procedures (e.g., different thermal treatment temperatures, different aging times, etc.), only the median is preserved as a more reliable representation of the polymer gas permeability. Eventually, a total number of 836 homopolymer structures are left in our training dataset, each with at least one permeability record for one of the five major industrial gases. For each gas, the total number of labeled data after cleaning is (H_2, 509), (O_2, 805), (N_2, 794), (CO_2, 754), and (CH_4, 681). Label distributions for the five gases of interest are shown in Figure S1. As we can see, the five datasets cover an ideally wide range of gas permeability values, however, with highly imbalanced distributions. This will be detrimental to training a model that can accurately identify high-performance polymer membranes because the target polymers always fall in regions rarely represented by the training data (e.g., two tail regions in the distributions corresponding to either high or low permeabilities).

The large unlabeled dataset is from PoLyInfo[S6], which is by far the largest experimental polymer database on polymeric materials design. It collects detailed information from literature on chemical structures, synthesizing and processing methods of samples, etc., covering a wide range of polymer subtypes, from homopolymers, copolymers, and polymer blends, to polymeric composites and compounds. In this work, only homopolymer is considered for the present study, which gives us a chemical space of 12,769 candidates for screening with guaranteed synthesizability by learning from the synthesizing procedures reported in the recorded literature.

Section 3. SAscores calculation

The synthetic accessibility score (SAscore)[S7] is calculated for the polymers in the training dataset with above-the-bound O_2/N_2 separation performance and compared with the SAscores of P130093 and P432352 (see Figure S11b). The source code for the calculation of SAscore as described in Ref. [S7] with several small modifications was adopted and can be found in Ref. [S8].

Table S1. Performance of models using only the labeled dataset for the prediction of five gas permeability tasks. Average MAE values with standard deviations on the validation dataset are shown. The best performance across all models for each task is bolded. All MAE values are in units of $\log_{10}$Barrer, where 1 Barrer = 10^{-10}cm^3(STP)cm/(cm^2 s cmHg).

GNN-based model	*Gas*	*N_2*	*O_2*	*H_2*	*CH_4*	*CO_2*
Yes	**GREA-L**	**0.448±0.108**	**0.382±0.030**	**0.271±0.033**	**0.402±0.060**	**0.348±0.112**
	GCN-L	0.604±0.255	0.434±0.068	0.300±0.092	0.475±0.071	0.374±0.066
	GIN-L	0.608±0.078	0.488±0.088	0.359±0.080	0.567±0.179	0.368±0.066
No	**RF**	0.556±0.011	0.528±0.010	0.493±0.010	0.646±0.011	0.749±0.013
	GPR	1.059±0.000	1.036±0.000	1.084±0.000	0.923±0.000	1.167±0.000
	MLP	0.529±0.020	0.473±0.022	0.558±0.020	0.515±0.025	0.861±0.027

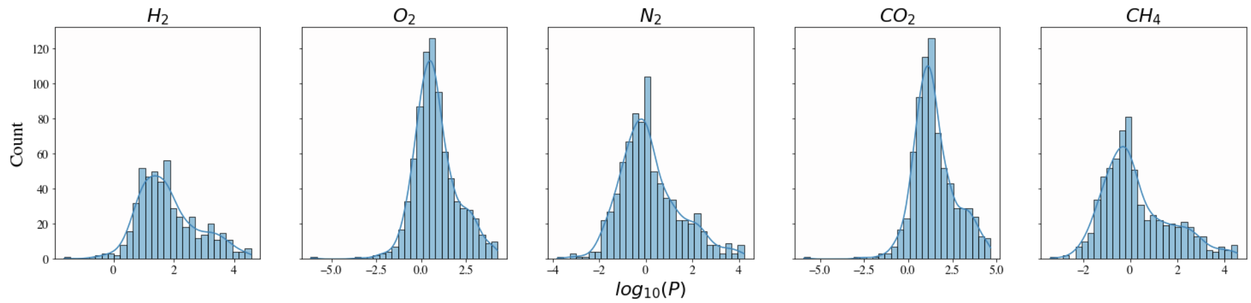

Figure S1. Label distribution of available permeability data for five major gases: H_2, O_2, N_2, CO_2, and CH_4. Permeability (P) is in units of Barrer (1 Barrer = 1×10^{-10} cm^3 [STP] cm^2/cm^3 s cmHg, where STP is the Standard Temperature Pressure, 1 bar and 273 K). The x-axis is in the unit of log_{10}Barrer.

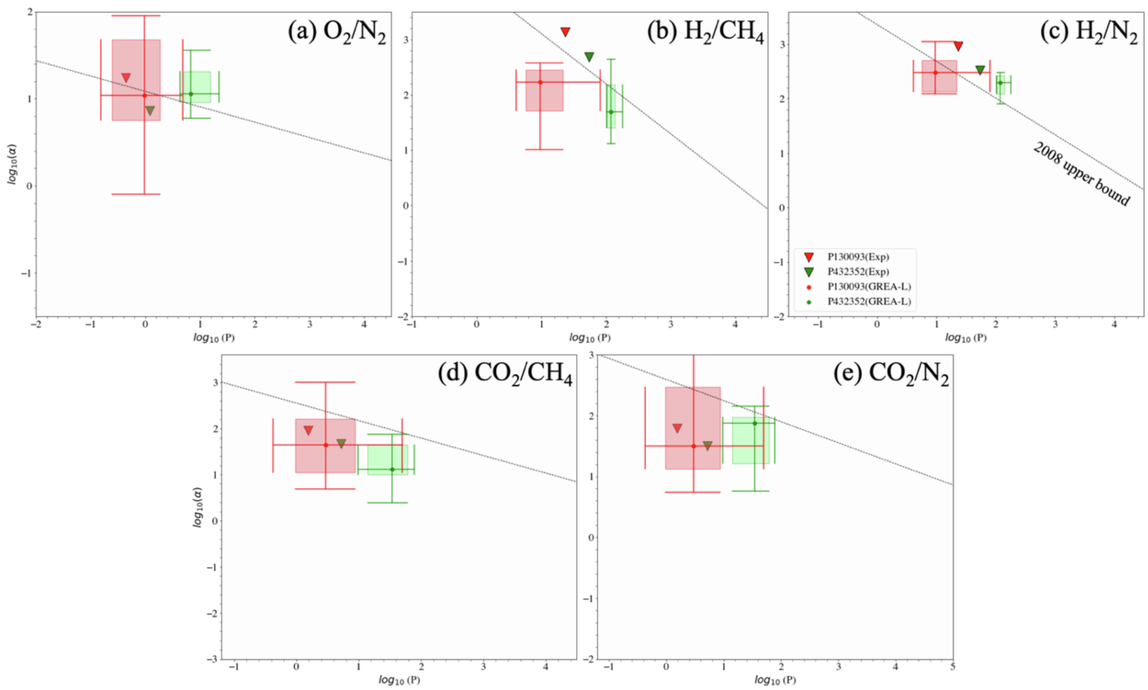

Figure S2. Robeson plot visualization for five different separation tasks using experiment values and GREA-L predictions.

Accuracy=5/10.

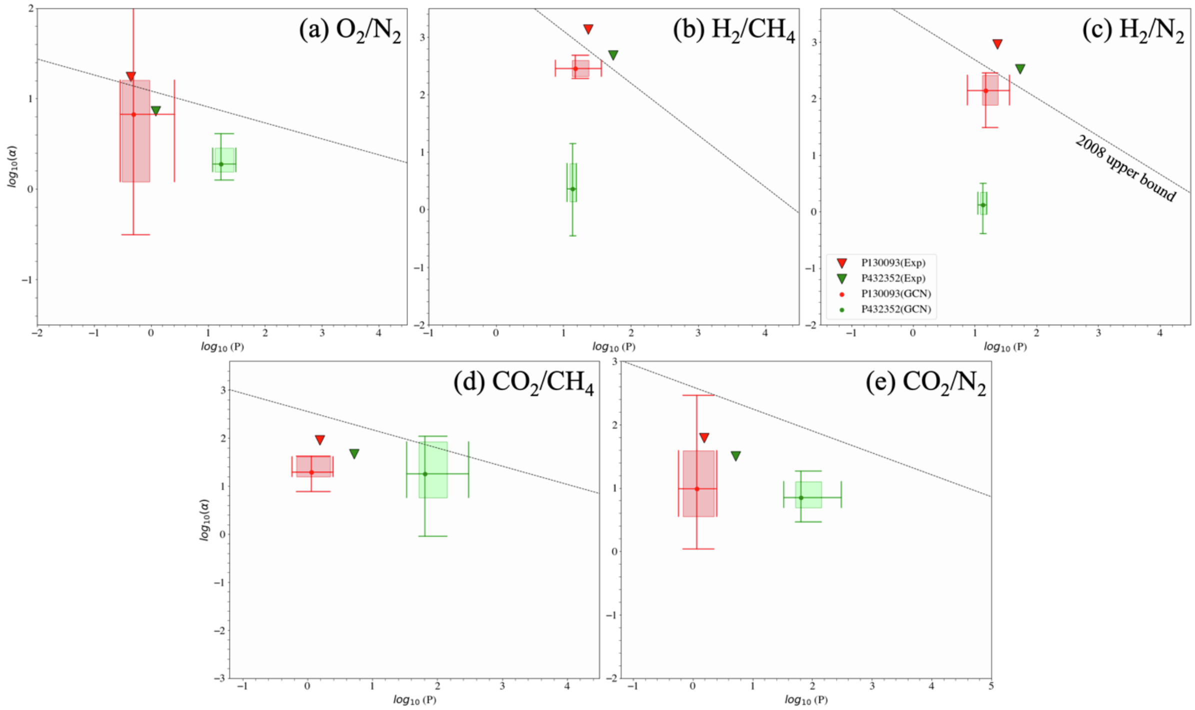

Figure S3. Robeson plot visualization for five different separation tasks using experiment values and GCN predictions. Accuracy=5/10.

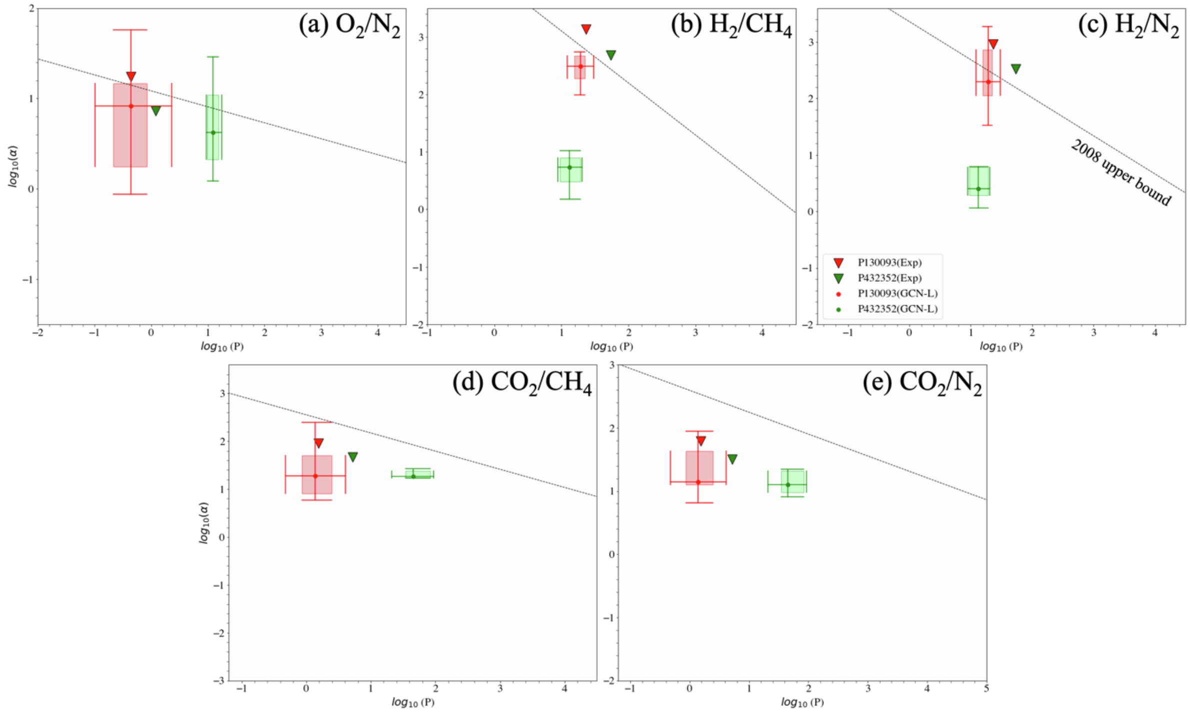

Figure S4. Robeson plot visualization for five different separation tasks using experiment values and GCN-L predictions.

Accuracy=5/10.

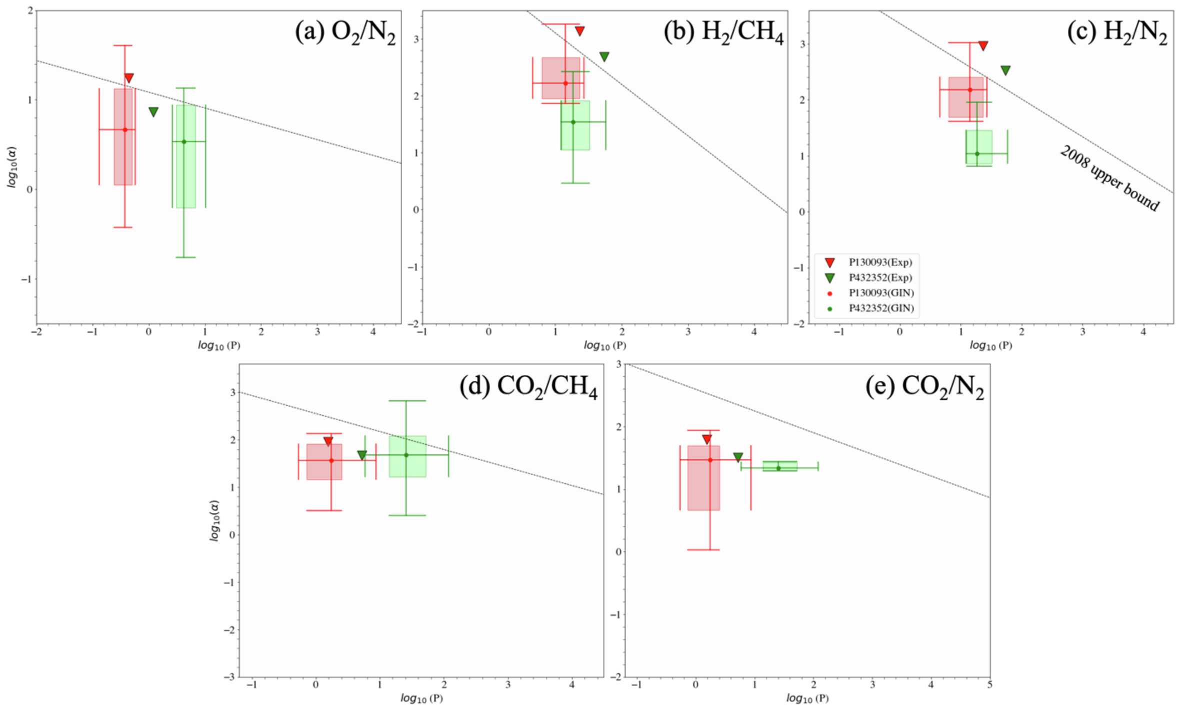

Figure S5. Robeson plot visualization for five different separation tasks using experiment values and GIN predictions. Accuracy=5/10.

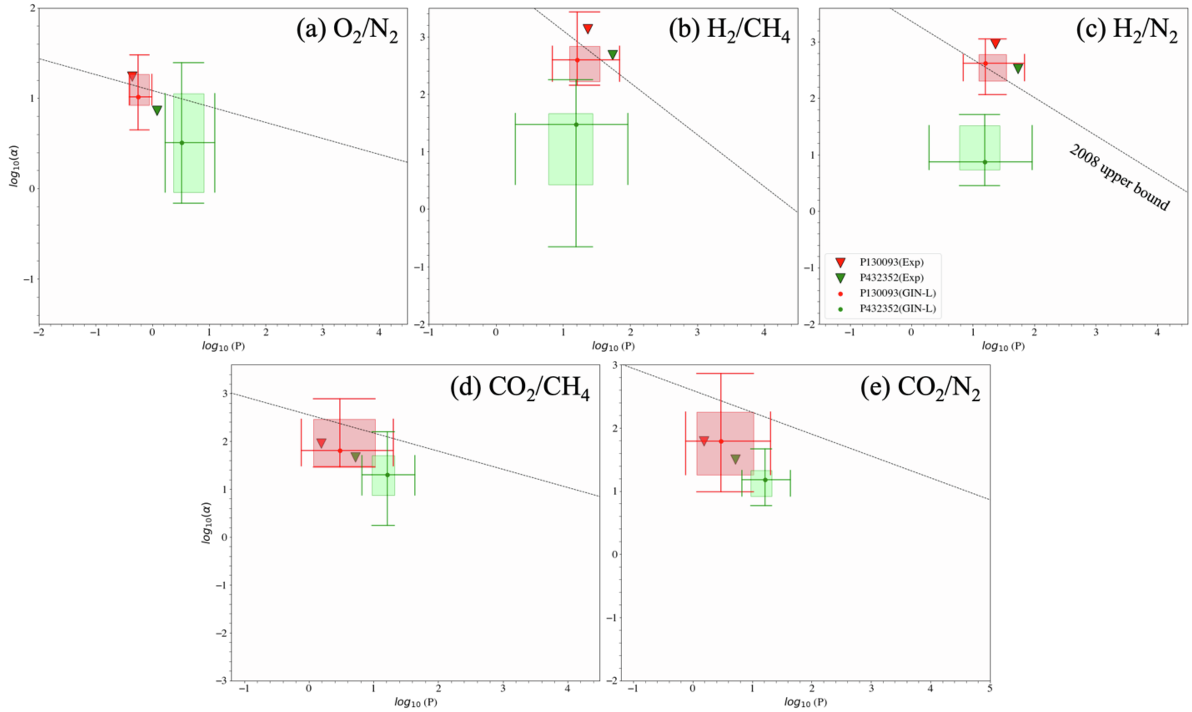

Figure S6. Robeson plot visualization for five different separation tasks using experiment values and GIN-L predictions.

Accuracy=6/10.

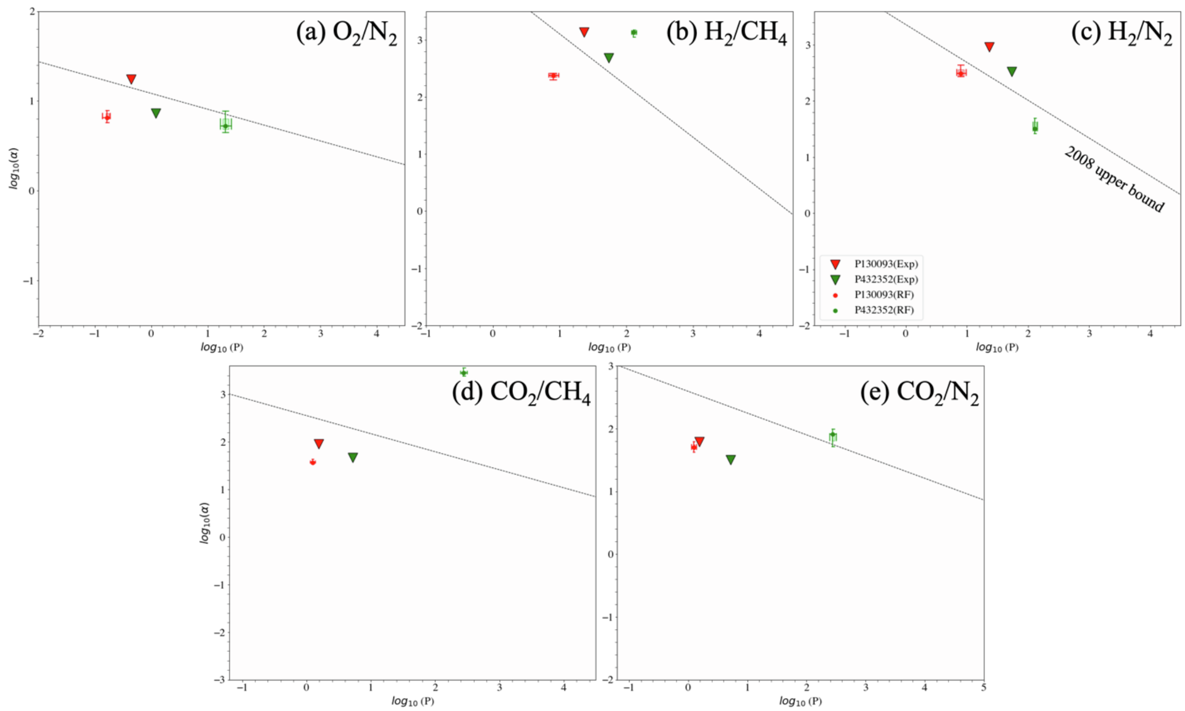

Figure S7. Robeson plot visualization for five different separation tasks using experiment values and RF predictions. Accuracy=4/10.

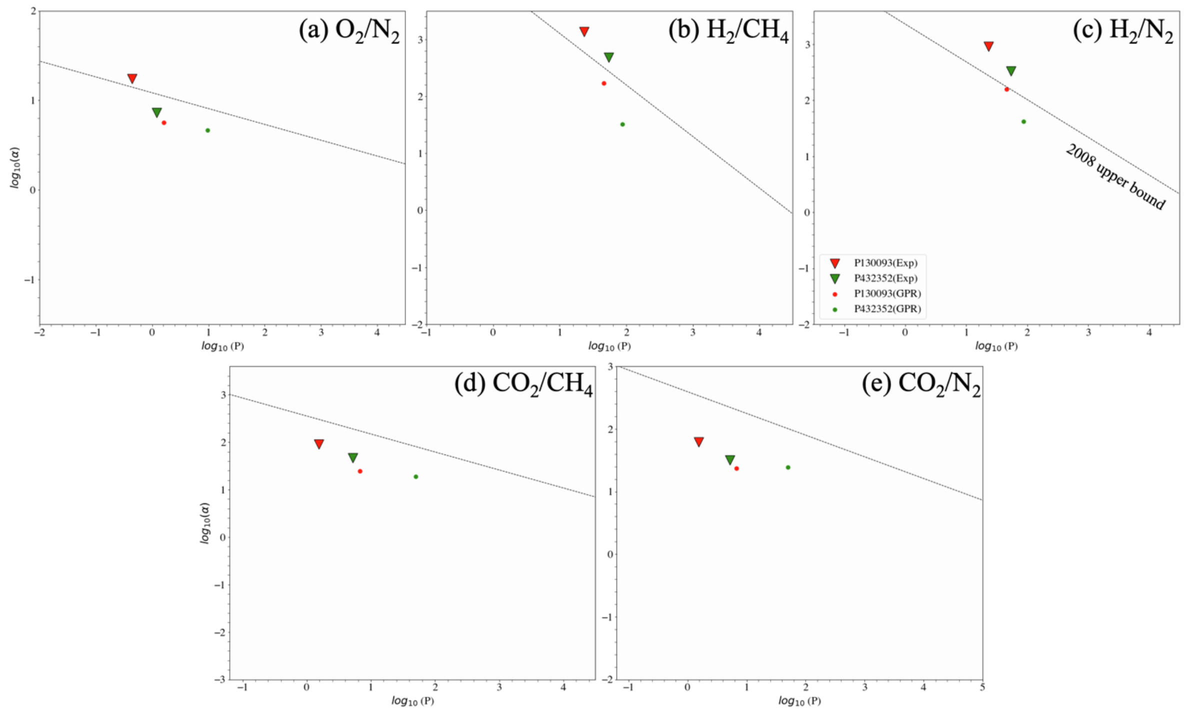

Figure S8. Robeson plot visualization for five different separation tasks using experiment values and GPR predictions. Accuracy=5/10.

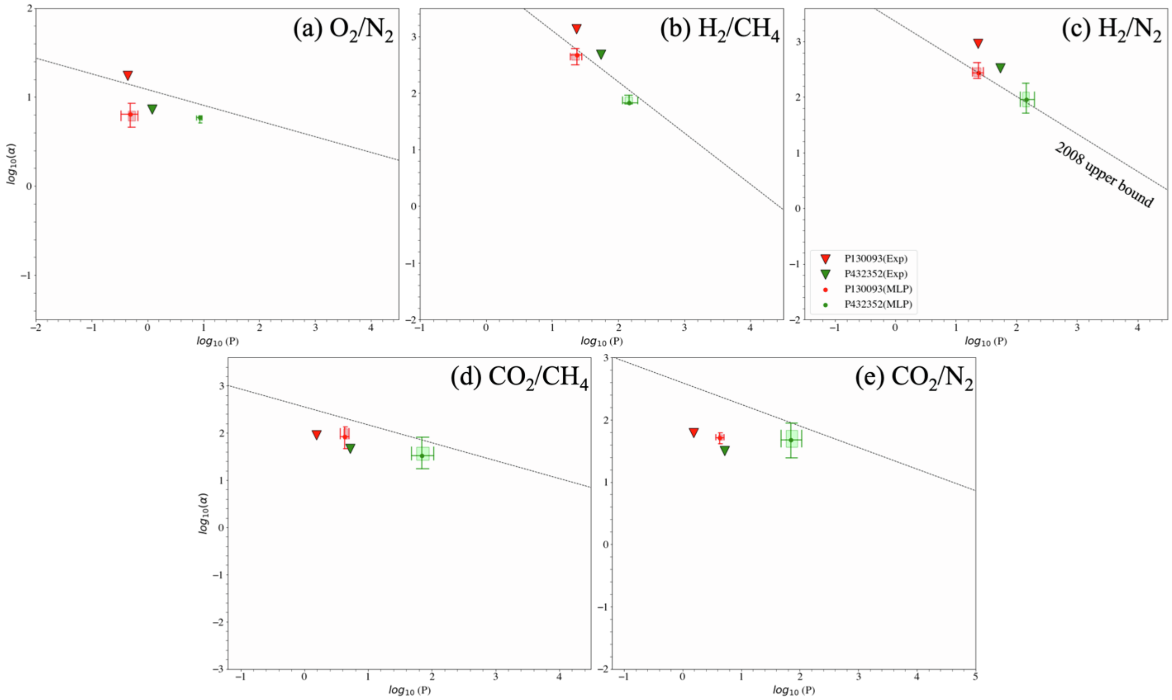

Figure S9. Robeson plot visualization for five different separation tasks using experiment values and MLP predictions. Accuracy=6/10.

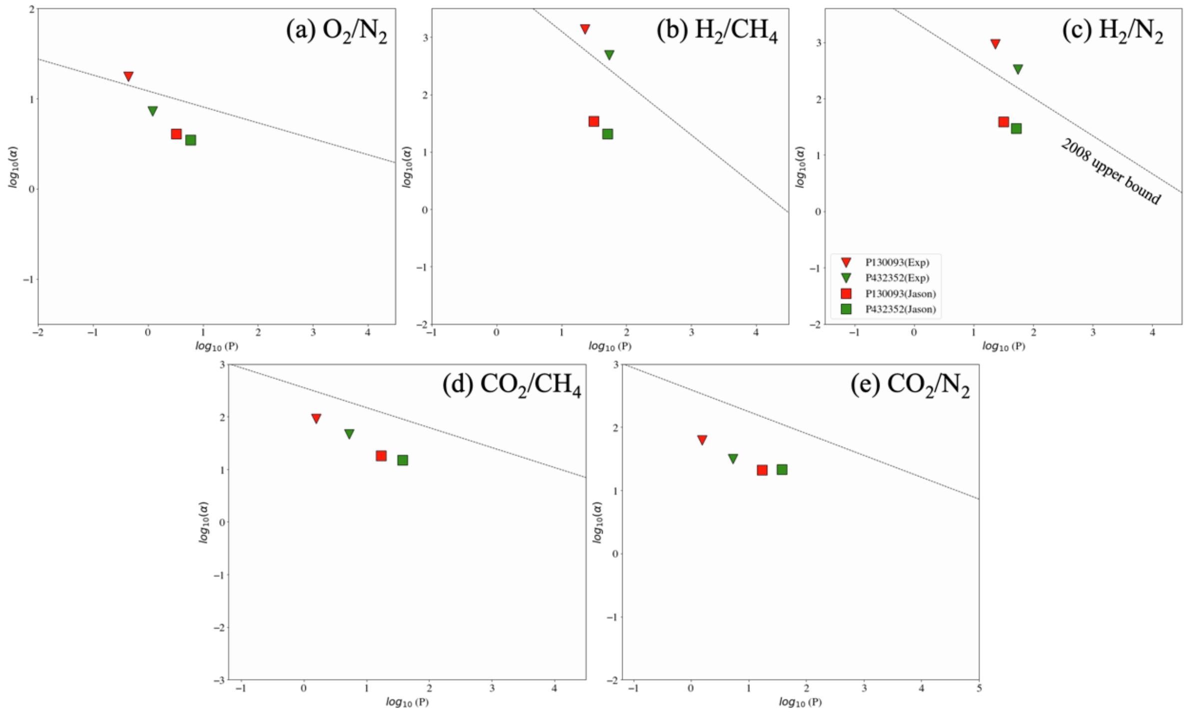

Figure S10. Robeson plot visualization for five different separation tasks using experiment values and predictions from Jason Yang's models. Accuracy=5/10.

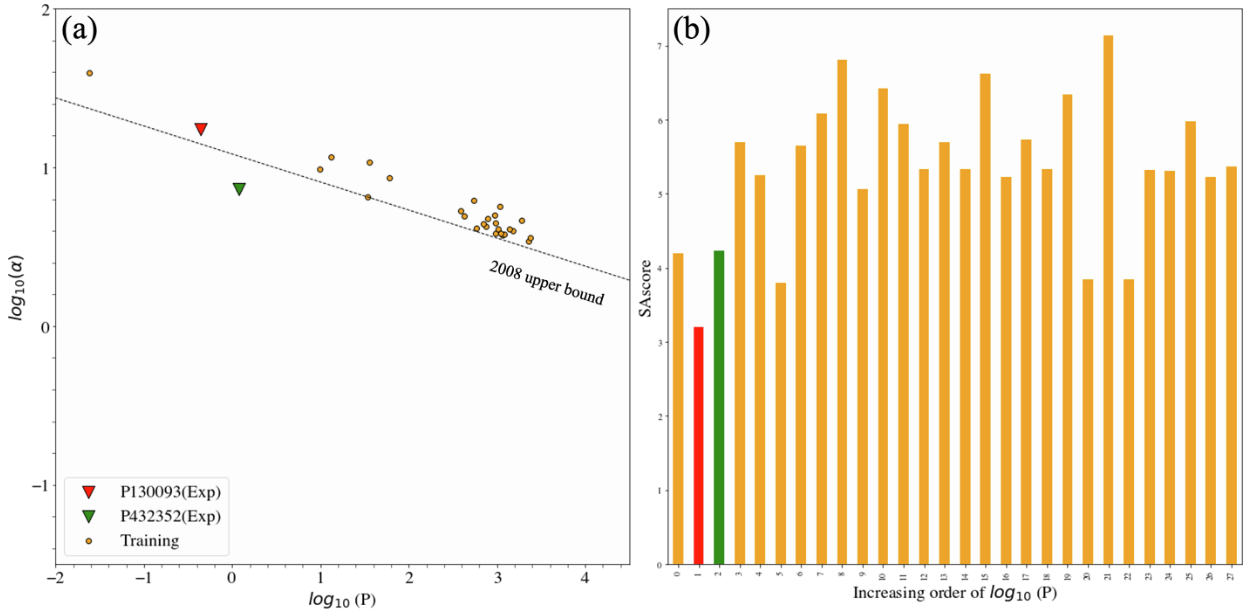

Figure S11. **Polymers in the training dataset with above-the-bound O_2/N_2 separation performance vs. the two synthesized polymers**. **(a)** Robeson plot visualization for O_2/N_2 separation; **(b)** SAscore of polymers along the increasing order of $\log_{10}P$ (left to right). Color orange: training data; red: P130093; green: P432352.

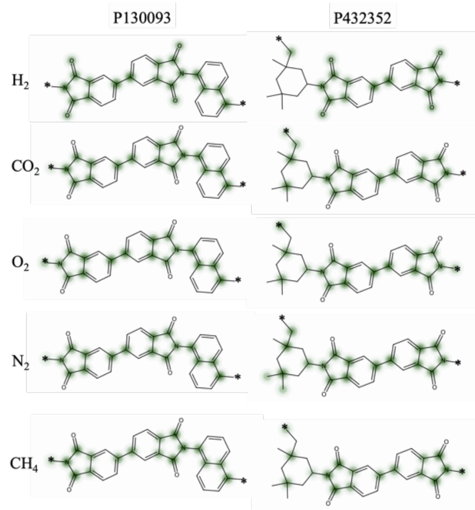

Figure S12. **Molecular graph rationales visualization for P130093 (left column) and P432352 (right column) on five permeability prediction tasks**, including H_2, O_2, N_2, CO_2, and CH_4 (top to bottom). We visualize the rationale subgraphs using the averaged rationale scores (from 10 independent runs) among the top 50%. The rationale subgraphs are highlighted in green. The darker the color, the higher the rationale score.

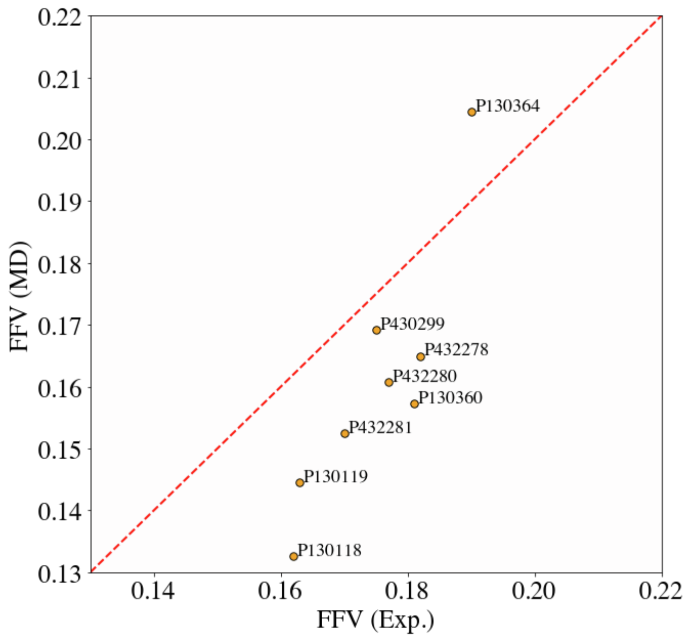

Figure S13. Benchmark of MD simulated FFV values of 8 polymers from literature [S9]. Polymers are annotated by their PIDs from PoLyInfo. The MD simulated FFV well captures the trend of experimentally measured FFV.

Table S2. Comparison of FFV reported in the literature (FFV-Exp) [S9] and FFV calculated by MD (FFV-MD) in this work. Polymers are annotated by their PIDs from PoLyInfo. The FFV-MD is averaged across the five different snapshots as described in the Methods section. The mean and standard deviation are reported for FFV-MD.

PID	FFV-Exp	FFV-MD
P432283	0.19	0.141 ± 0.041
P130364	0.19	0.204 ± 0.009
P430299	0.175	0.169 ± 0.028
P432278	0.182	0.165 ± 0.021
P432281	0.17	0.153 ± 0.017
P432280	0.177	0.161 ± 0.012
P130360	0.181	0.157 ± 0.026
P130118	0.162	0.133 ± 0.022
P130119	0.163	0.145 ± 0.042

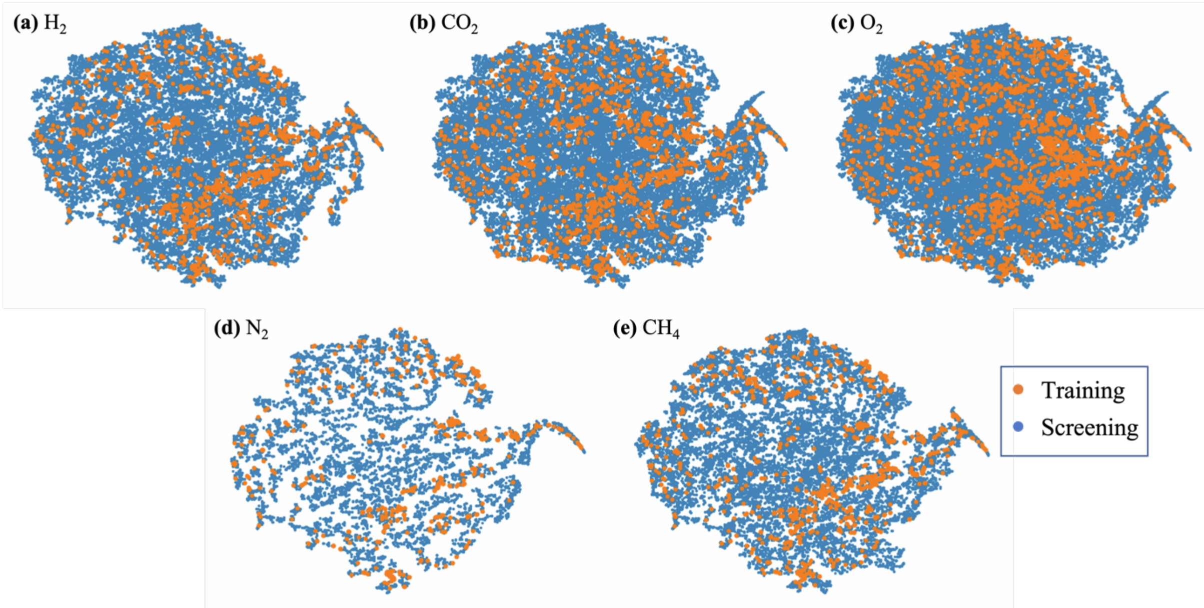

Figure S14. The t-SNE[S10] graph embedding visualization for training and screening datasets on five permeability prediction tasks, including (a) H_2, (b) CO_2, (c) O_2, (d) N_2, and (e) CH_4. The screening dataset (PoLyInfo) generally spans the feature space of the training data on five different tasks.

Table S3. Gas permeation test of polymer P130093. $\left(\frac{dp}{dt}\right)_{ss}$ and $\left(\frac{dp}{dt}\right)_{leak}$ are the steady-state pressure increment in downstream, and the leak rate of the system (cmHg/s), respectively. l is the thickness of the polymer film and A is the effective film area. Measurement errors are shown in parentheses. H_2 is not included in the diffusivity calculation because it has a very short lag time and cannot be detected by our existing techniques. $\frac{dp}{dt}$ has a unit of torr/s.

		H_2	CH_4	N_2	O_2	CO_2
$\left(\frac{dp}{dt}\right)_{leak}$		2.97E-06	1.21E-06	1.47E-06	1.75E-06	3.74E-07
$\left(\frac{dp}{dt}\right)_{ss}$	30 psig	1.26E-05	1.30E-06	1.87E-06	9.12E-06	3.90E-06
	80 psig	2.43E-05	1.74E-06	2.19E-06	1.74E-05	7.72E-06
	130 psig	3.66E-05	2.15E-06	2.85E-06	2.55E-05	1.09E-05
Lag-time (s)		\	2667.42 (440.03)	2508.69 (298.42)	1281.12 (60.83)	2385.24 (292.07)
l (cm)		0.00663 (0.00008)				
A (cm^2)		0.2391 (0.0006)				

Table S4. Gas permeation test of polymer P432352. $\left(\frac{dp}{dt}\right)_{ss}$ and $\left(\frac{dp}{dt}\right)_{leak}$ are the steady-state pressure increment in downstream, and the leak rate of the system (cmHg/s), respectively. l is the thickness of the polymer film and A is the effective film area. Measurement errors are shown in parentheses. H_2 is not included in the diffusivity calculation because it has a very short lag time and cannot be detected by our existing techniques. The $\left(\frac{dp}{dt}\right)_{ss}$ of CH_4 and N_2 at *50 psig* are skipped because of the very slow permeation. $\frac{dp}{dt}$ has a unit of torr/s.

		H_2	CH_4	N_2	O_2	CO_2
$\left(\frac{dp}{dt}\right)_{leak}$		1.20E-06	2.13E-06	1.89E-06	1.72E-06	4.19E-07
$\left(\frac{dp}{dt}\right)_{ss}$	30 psig	3.13E-05	4.55E-06	5.34E-06	2.73E-05	1.48E-05
	50 psig	4.43E-05	\	\	3.85E-05	2.10E-05
	80 psig	6.52E-05	7.21E-06	9.33E-06	5.56E-05	2.99E-05
Lag-time (s)		\	2320.96 (123.01)	606.04 (110.46)	114.84 (20.21)	490.12 (36.73)
l (cm)		0.00408 (0.000056)				
Area (cm^2)		0.1861 (0.00065)				

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